

Aparna Ramesh Pai

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Education

Stockholm University — M.Sc. Data Science, Statistics & Decision Analysis *Sep 2024 – Jun 2026*
Stockholm, Sweden

B.M.S. College of Engineering — B.Tech. Artificial Intelligence & Machine Learning *Oct 2020 – Jun 2024*
Bengaluru, India

Experience

Data Science Intern — Aditya Auto Products & Engineering Pvt. Ltd., Bengaluru *Jun 2025 – Aug 2025*

- Identified gap in reactive maintenance practices at a power press manufacturing facility; independently designed and prototyped a predictive maintenance system, visiting the site to observe machines and generating synthetic training data in the absence of historical records
- Built and validated a LightGBM fault-detection model achieving **86% classification accuracy** on a heavily imbalanced dataset through custom class-weighting and iterative feature engineering
- Prototype and proposed architecture were compelling enough for the company to greenlight further implementation — delivering a tangible business outcome within a 2-month engagement at a resource-constrained SME

Data Analyst Intern — Bosch, Bengaluru *Nov 2023 – Apr 2024*

- Conducted end-to-end baseload energy analysis across Bosch's Adugodi campus for a green energy transition initiative — integrating and preprocessing **1 year of raw consumption data** from multiple sources including building sensors, occupancy logs, weather patterns, and seasonal records
- Identified peak consumption hours, mandatory-load requirements, and **24/7 electricity waste in non-automated zones** by cross-referencing building occupancy, working vs. non-working days, and unautomated facility areas — producing a structured per-building consumption breakdown
- Delivered **data-backed automation recommendations** across 5 campus buildings, identifying 2–3 zones with significant 24/7 electricity waste as priority automation candidates — directly supporting Bosch's sustainability roadmap using Power BI and analytics platforms

Machine Learning Intern — Indian Institute of Science (IISc), Bengaluru *Feb 2023 – Jun 2023*

- Developed neural network model to predict streamflow at the Netravati River outlet for a climate & hydrology research project, achieving **92% prediction accuracy** through iterative model tuning and feature engineering
- Conducted end-to-end data analysis pipeline including feature selection, performance evaluation, and result visualisation to support PhD-level research publication

Projects

Master's Thesis — Flexibility of Nuclear Plants in the Nordic Electricity Market *2025 – 2026*

- Modelling operational flexibility of nuclear assets within the Nordic grid under variable renewable penetration scenarios, with direct implications for energy transition strategy in Sweden

FinChurn — End-to-End Customer Churn Prediction System *2025*

- Built production-grade churn prediction pipeline using XGBoost, FastAPI, MLflow experiment tracking, and Streamlit dashboard; designed for real-time inference and A/B testing readiness

HOLESON — Uncertainty-Aware Black Hole Parameter Inference via SOMs *2026*

- Applied Self-Organising Maps to astrophysical datasets for unsupervised parameter estimation with quantified uncertainty bounds

Technical Skills

Programming: Python, R, SQL, GAMS Studio

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, LightGBM, XGBoost, TensorFlow, FastAPI, MLflow, Streamlit, Power BI, n8n

ML Techniques: Supervised & Unsupervised Learning, Neural Networks, LLM Integration, Time-Series Forecasting, Media Mix Modelling

Languages: English (Fluent), Korean (Intermediate), Swedish (Beginner)